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2.2.7 Why we love unitary matrices

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A could be pretty bad. If A is almost singular then it would be bad. Now as the

course goes on, we're going to see a lot of algorithms that have the property

that you apply a bunch of matrices to a vector or to a matrix in an effort to

transform those matrices into something nicer. Now every time you multiply by a

matrix you have the potential of amplifying relative error. That's bad.

What if we restricted ourselves to matrices that have a condition number

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Reading assignment

1/1 point (graded)



Read Unit 2.2.7 of the notes. [\[LINK\]](#)

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Homework 2.2.7.1

1/1 point (graded)
Let $\mathbf{A} \in \mathbb{C}^{n \times n}$ be nonsingular and $\mathbf{x} \in \mathbb{C}^n$ a nonzero vector. Consider

$\mathbf{y} = \mathbf{A}\mathbf{x} \quad \text{and} \quad \mathbf{y} + \delta \mathbf{y} = \mathbf{A}(\mathbf{x} + \delta \mathbf{x}).$

Show that

$$\frac{\|\delta \mathbf{y}\|}{\|\mathbf{y}\|} \leq \underbrace{\|\mathbf{A}\| \|\mathbf{A}^{-1}\|}_{\kappa(\mathbf{A})} \frac{\|\delta \mathbf{x}\|}{\|\mathbf{x}\|},$$

where $\|\cdot\|$ is an induced matrix norm.

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Solution:

For complete solution, see [[LINK](#)].

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 Answers are displayed within the problem

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